

NetMamba+

from source to a measured demo

Actual GPU training, saved-classifier inference and a practical IDS roadmap

3 × 120

completed fine-tuning epochs

86.65%

mean test accuracy · three seeds

1,041

flows in the fixed test split

Source-based fine-tuning on a GB10 compatibility port. Exact paper reproduction remains unestablished.

From an untested harness to GPU evidence

The work includes failures, fixes and a repeatable installation rehearsal.

Runtime

Built the authors' Mamba fork and causal convolution for GB10.

Seven forward/backward checks passed in two isolated environments.

Training

Masked pretraining: 132 updates on training-only flows.

Fine-tuning: three seeds, 120 epochs and 7,920 updates each.

Inference

Strictly reloaded the selected classifiers.

Recomputed confusion matrices and metrics from saved predictions.

Built an offline replay.

Commands, hashes, losses and prediction records are published alongside the explanation.

One prediction describes one flow

The uploaded packet CSVs cannot establish this ordered, multimodal flow representation.

Byte content

First 5 packets × 320 bytes
Tensor: [B, 1, 1, 1600]
400 embedded byte strides

Packet sizes

First 20 values
Native clipping and padding
20 size tokens

Arrival intervals

First 20 values
Native monotonic transform
20 interval tokens

443 tokens → four Mamba blocks → three summed summary vectors → six class logits

Three stages, three different questions

A pretraining checkpoint becomes a classifier only after supervised fine-tuning.

1 · Pretrain

Hide byte, size and timing entries.

Reconstruct the missing content.

Learn an encoder without attack-category targets.

2 · Fine-tune

Load the pretrained encoder.

Learn six labeled categories.

Select the checkpoint using validation accuracy.

3 · Infer

Freeze the selected classifier.

Load weights and class meanings strictly.

Emit six scores and a predicted label.

The released pretraining file has no classification head. Our saved fine-tuned classifier does.

The source recipe differs from the paper

Preserve those differences when describing the result.

Setting	Measured source profile	Paper v1
Fine-tuning budget	120 epochs × 3 seeds	120 epochs
Batch size	128	64
Maximum effective LR	0.001 (blr 0.002)	Reported 0.002
Runtime	GB10 / Torch 2.9.1 / cu130	A100 / Torch 2.1.1
Pretraining history	Released weights; partly unknown	Browser + Kitsune

Official splits: 8,323 train / 1,040 validation / 1,041 test. Seed 0 was chosen for the demo before test scores.

Report every seed and name the metric

Test accuracy, support-weighted F1 and macro F1 are independently recomputed.

Seed	Accuracy	Weighted F1	Macro F1
0	91.26%	91.26%	91.58%
1	84.05%	84.08%	84.86%
2	84.63%	84.67%	85.38%
Mean	86.65%	86.67%	87.27%

Accuracy sample SD: 4.00 percentage points across seeds on this fixed split.

Paper Table IV: NetMamba+ CICIoT2022 accuracy/F1 = 97.50%. This is a reference under different conditions.

Make the errors visible

Seed 0, selected for the demo in advance. Rows are true classes; columns are predictions.

TRUE / PRED.	Flood	RTSP	Audio	Other	Camera	Home
Flood	185	14	0	0	1	0
RTSP	1	199	0	0	0	0
Audio	0	0	185	1	12	2
Other	0	1	1	37	0	2
Camera	0	0	7	0	168	25
Home	1	1	6	0	16	176

Seed 0 accuracy

91.26%

Seed 0 macro F1

91.58%

Test examples

1,041 flows / six classes

The smallest test class contains 41 flows. Keep per-class support and recall beside the overall score.

Replay actual saved-classifier predictions

The browser page works offline. GPU inference produced the records behind it.

NETMAMBA+ / MEASURED SYSTEM WALKTHROUGH

Recorded inference replay

Follow a saved classifier as it labels held-out CICIoT2022 flows. Every row below comes from a real GPU inference run.

Offline demonstration
This is not live packet capture. Playback speed controls the display, not the model. No traffic is blocked.

Input · bytes + sizes + timing → NetMamba+ · 443 tokens / flow → Output · six class scores

1,041 / 1,041
flows displayed

91.26%
full test accuracy

91.58%
full test macro F1

402
displayed attack-class predictions

[Play replay](#)
[Reset](#)
[Show all](#)
 Display pace: Fast · 150 ms / row
 Rows: All displayed flows

ROW	MODEL PREDICTION	SCORE	DATASET LABEL	MATCH
1041	Home Automation	92.29%	Home Automation	Correct
1040	Home Automation	91.49%	Home Automation	Correct
1039	Home Automation	89.78%	Home Automation	Correct
1038	Cameras	54.62%	Home Automation	Error
1037	Home Automation	91.42%	Home Automation	Correct
1036	Home Automation	92.04%	Home Automation	Correct
1035	Home Automation	86.64%	Home Automation	Correct

Predictions and benchmark truth
Replay complete · 1041 rows match the selected filter.

How to read this

- Prediction is the class with the largest model logit.
- Score is uncalibrated softmax confidence; it is not a measured probability of malicious traffic.
- Dataset label is benchmark truth used for evaluation. It is not a model input.
- Attack-class means Flood or RTSP Brute Force in this six-class benchmark. Other attacks are not covered.
- Errors are retained. Official splits contain a small number of exact raw-input overlaps.

CLASSIFIER FINGERPRINT
b7f9a3a01d2d14cc9f61b4c5166fba335b48c

Present it

Play, then Show all.

Filter Prediction errors.

Explain one mistake and show the checkpoint hash.

State the limit

Recorded inference only.

No capture or blocking.

Display pace is adjustable.

Scores are uncalibrated.

A recorded walkthrough and PDF are included as meeting backups.

Time the model and state what is excluded

GB10 · float16 autocast · 20 warmups + 100 synchronized measurements per batch size

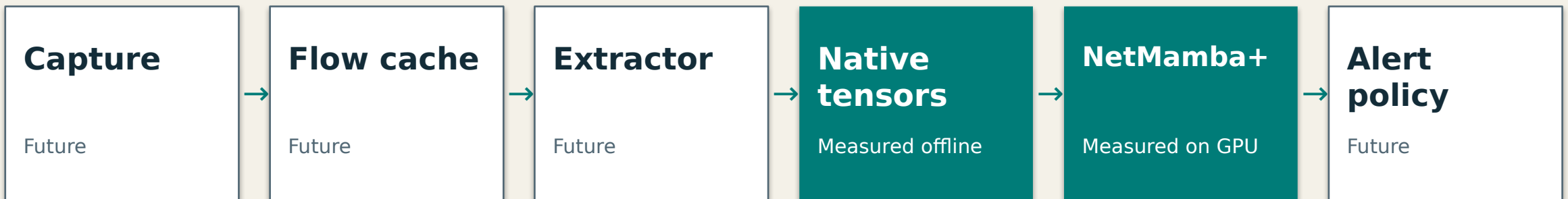
Batch	Median batch latency	95th percentile	Flows/s from mean
1	0.88 ms	0.93 ms	1,120.0
16	4.32 ms	4.64 ms	3,679.1
128	36.55 ms	37.38 ms	3,502.3

First 128 flows: 100.00% prediction agreement between single-flow and batched execution.

These are model-only flows/second. Capture, flow waiting, preprocessing, transfers and alert handling are excluded.

The next build is a capture-to-alert system

The classifier is measured. The online integration still needs its own implementation and tests.



Measure independent-traffic accuracy, false alerts, packet loss, flow waiting and queue delay.

First milestone: a reviewed PCAP produces the same tensors as the training extractor, then run alert-only trials.

Split packet handling from neural inference

No NPU or SmartNIC execution is claimed in this project.

SmartNIC / DPU

Steer, filter, count and queue packets.

Host or Arm cores manage flows and extraction.

DOCA Flow is a candidate integration layer.

GPU / AI NPU

GPU path works here.

An NPU needs a named compiler/device and support for the selective scan, causal convolution and normalization.

Proof before sizing

Export status: not captured in the tested Torch path

Then verify precision, memory and per-class accuracy.

Measure on the target hardware.

DGX Spark does not support GPUDirect RDMA in current DOCA GPUNetIO documentation. Do not assume a direct NIC-to-GPU path.

A measured research baseline to build from

We trained and tested the original NetMamba+ classifier on compatible flows, with a documented GB10 runtime port.

Ready to show

- Three completed fine-tuning runs
- Strict saved-model inference
- Independent metrics and errors
- Offline replay, documents and evidence

Still to establish

- Full Browser/Kitsune pretraining history
- Exact paper-protocol reproduction
- Independent customer traffic + live extraction
- Calibrated IDS policy + target hardware execution

Next decision: agree on representative traffic and a specific deployment target before promising operational performance.

Keep the model; add a verified execution route

The paper proposes a method. The authors release research code and assets. This repo records a measured port.

Authors' research

Original NetMamba+ model
Native loader and engine
Compatible flow data
Pretrained encoder weights

Our additions

Source and asset checks
GB10 build support
Strict evaluation +
prediction
Replay, results and guides

Disclosed changes

Compiler edits in build
copies
New GPU / Torch runtime
Source batch and rate
recipe
Unknown pretraining history

The original model and loader checkout is unchanged. Runtime and experiment differences prevent an exact paper reproduction claim.

Choose the route that matches your goal

Viewing, verifying and running the model have different requirements.

Present now

Open the guide and recorded replay in a browser.

Download the ZIP for offline use.

No Python or GPU needed.

Check the package

Clone or unzip the project.

Use Python 3.10 or newer.

Run the CPU tests and package verifier.

No training assets needed.

Run new inference

Follow the GB10 setup runbook.

Fetch pinned data and source.

Train a local classifier, then evaluate or predict.

Start at [docs/customer/quickstart.md](#). The full GPU setup is tested on GB10, not promised for every computer.

Read a passing test in its proper scope

Passing software checks and successful GPU runs answer different questions.

CPU checks

54 tests passed.

Saved metrics, hashes, links
and document alignment
checked.

Prior package CI passed on
Python 3.10 and 3.12.

Fresh GPU checks

All three model-only exports
reproduced their scores.

Unlabeled inference
matched all 1,041 seed-0
predictions in this run.

Known limit

GPU inference works.

The tested Torch export
path cannot trace a custom
CUDA operation.

No NPU or live IDS test has
passed.

See [verification.md](#) for dated evidence and [quickstart.md](#) for commands. A passing CPU test does not rerun GPU training.

Every measured claim has a saved artifact

The repository is the handoff: explanation, editable files, commands and reviewed evidence.

Start here

docs/customer/README.md
Briefing PDF + PowerPoint +
slide PDF
Runbook + talk track +
recorded demo

Verify results

evidence/results.json
Per-seed manifests, epoch
logs and predictions
Runtime parity + build
records + benchmark

Read the research

docs/research/research-
record.md
Pinned paper/source links
Exact council decision and its
evidence limits

github.com/buffbeefalo/netmambaplus-reproduction